AGU PUBLICATIONS

JGR

Journal of Geophysical Research: Solid Earth

RESEARCH ARTICLE
10.1002/2015JB012577

Key Points:
- One of the largest impacts on Earth provoked no disturbance of the geodynamo
- Magnetic anisotropy of impact melt indicates radial flow during crater formation
- Titanohematite in the impact melt formed via degassing and oxygen diffusion

Supporting Information:
- Tables S1 and S2

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Citation:
Eitel, M., S. A. Gilder, J. Spray, L. Thompson, and J. Pohl (2016), A paleomagnetic and rock magnetic study of the Manicouagan impact structure: Implications for crater formation and geodynamo effects, *J. Geophys. Res. Solid Earth*, *121*, 436–454, doi:10.1002/2015JB012577.

Received 2 OCT 2015
Accepted 31 DEC 2015
Accepted article online 5 JAN 2016
Published online 1 FEB 2016

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A paleomagnetic and rock magnetic study of the Manicouagan impact structure: Implications for crater formation and geodynamo effects

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Abstract We report rock magnetic and paleomagnetic data from the ~214 Ma Manicouagan (Canada) impact crater based on 25 widely distributed sites of impact melt and basement rocks collected at the surface as well as from boreholes drilled to depths ≤1.5 km. Titanomagnetite and titanohematite carry the magnetic remanence in impact melts above 320 m elevation and in most basement rocks. Impact melts below 320 m contain solely titanomagnetite. Magnetic susceptibility and saturation magnetization, proxies for titanomagnetite concentration, increase more than tenfold toward the base of the thickest impact melt that underwent fractional crystallization. The titanomagnetite-enriched zone partially contributes to a 2000 nT magnetic anomaly in the crater's center. Stepwise demagnetization reveals a single, normal polarity magnetization component in all samples regardless of the magnetic phases present. Coeval lock-in remanence times for titanomagnetite and titanohematite indicate that the titanohematite formed >570°C during oxi-exsolution. The average paleomagnetic direction and intensity coincide well with 214 Ma reference values. We find no evidence for an aberration of the geomagnetic field over the several thousands of years it took to cool a 481 m thick portion of the impact melt body. Hence, the energy released by the Manicouagan impact that created one of the 10 largest known craters on Earth provoked no measurable disturbance of the geodynamo. Magnetic anisotropy of clast-free impact melts define magnetic lineation directions that are, in places, radially oriented with respect to the crater's center. Centrifugal flow of the melt within the evolving transient crater probably generated the fabric.

1. Introduction

Meteorite impact craters are common morphologic features in the solar system. The energy released during impact events can exceed that of the largest terrestrial earthquakes by several orders of magnitude. Some of this kinetic energy penetrates deep within the planet, potentially influencing the magnetohydrodynamic regime in the core, which could in turn perturb the dynamo generation process (see *Eitel et al.* [2014] for a more detailed discussion). Evidence linking meteorite impacts to anomalous magnetic field behavior arises from correlating tektite layers with the paleomagnetic records in marine sediments [*Glass and Heezen*, 1967; *Durrani and Khan*, 1971; *Glass and Zwart*, 1979; *Hartl and Tauxe*, 1996]; however, the correlations are highly debated [*Burns*, 1990; *deMenocal et al.*, 1990; *Schneider and Kent*, 1990; *Glass et al.*, 1991; *Schneider et al.*, 1992].

To test the link between dynamo perturbation and high-energy shock events, we initiated a paleomagnetic study of rocks within meteorite craters whose ferromagnetic minerals cooled through their Curie temperatures after impact. The objective is to sample impact lithologies as a function of cooling time, e.g., from the margins toward the interior of impact melt sheets, to identify potential changes in magnetic field direction and/or intensity. To date we have carried out such work in the Ries (Germany [*Koch et al.*, 2012]), Rochechouart (France [*Eitel et al.*, 2014]), and Mistastin (Labrador, Canada [*Hervé et al.*, 2015]) craters. Anomalous field behavior has not been observed in any of them—possibly because the energy released from these impact events was insufficient to have an influence or because the rocks cooled too fast after impact and "missed" the event. The amount of time captured by the magnetic record in these craters did not exceed 400 years.

We now focus on the Manicouagan crater (Québec, Canada; Figure 1), whose rim-to-rim diameter of 85 km ranks it as one of the ten largest impact craters on Earth [*Earth Impact Database*, 2015]—2 to 3 times larger in diameter than the craters we previously studied. Estimates of the energy released from this event range

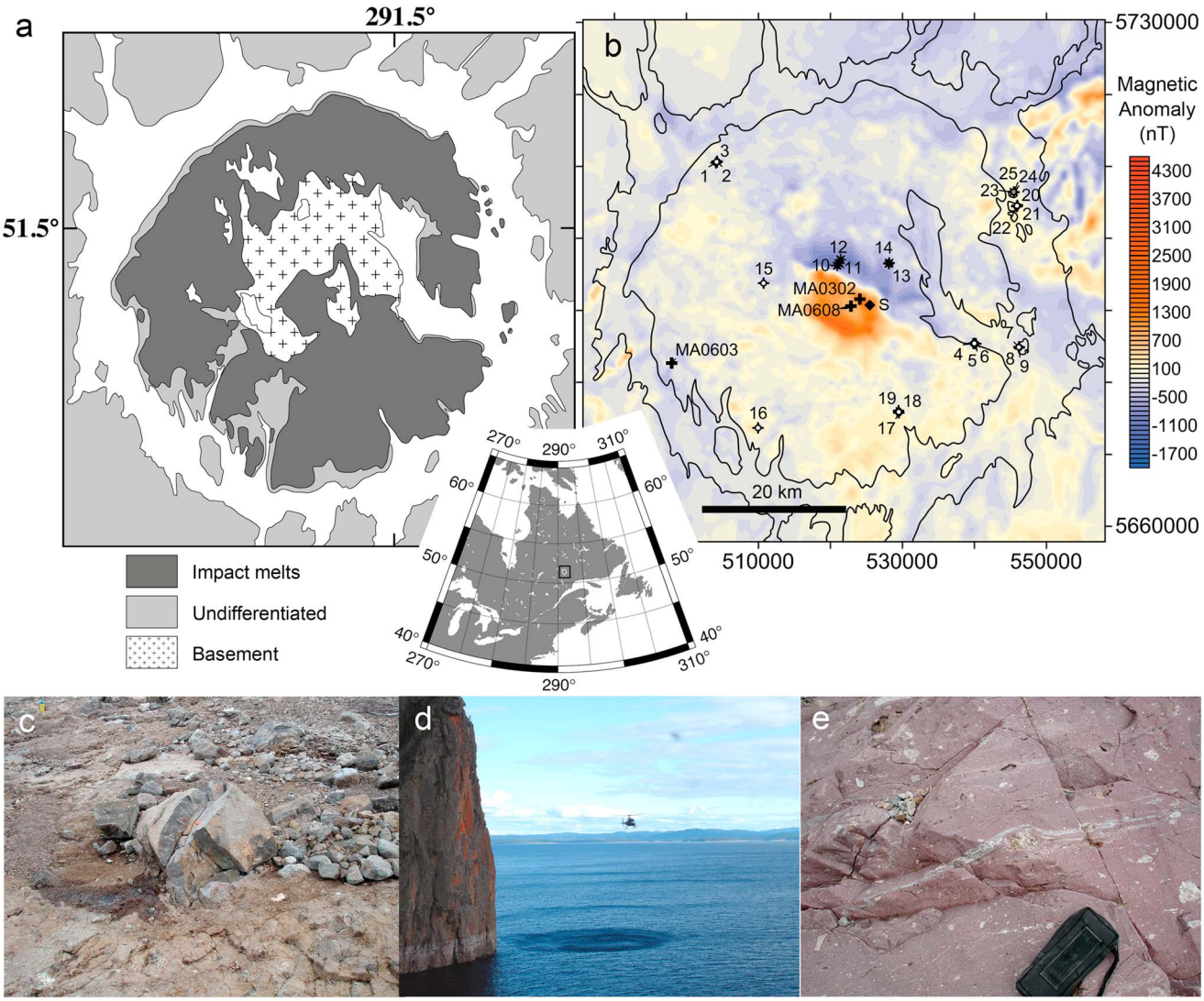


Figure 1. (a) Simplified geologic map of the Manicouagan impact crater [after *Spray et al*., 2010] with regional geographic location map (inset). (b) Magnetic anomaly map [*Natural Resources of Canada, Geoscience Data Repository for Geophysical Data*, 2014: http://gdr.agg.nrcan.gc.ca/gdrdap/dap/search-eng.php] including paleomagnetic sampling sites (numbered open symbols are impact melts, and filled symbols are basement rocks); drill cores used in this study are represented by pluses (e.g., MA0608); S is the drill core analyzed by *Scott et al*. [1996]. Field photos of (c) site 5 showing melt-bearing breccias with highly rounded basement clasts with diameters ranging from centimeters to meters; (d) impact melts near sites 8 and 9—annular lake and helicopter in view to south; and (e) basal, clast-laden impact melt near site 24 showing a flow fabric.

between 10^{22} and 10^{23} J [*Dence et al*., 1977; *Phinney and Simonds*, 1977]. Moreover, recent drilling results reveal melt thicknesses up to 1500 m at Manicouagan, which, except for Sudbury, exceed other known terrestrial impact melt bodies nearly tenfold [*Spray and Thompson*, 2008], potentially allowing us to sample a much longer record of field behavior following the impact event. Cooling rate calculations, which have inherent uncertainties, such as the original thickness and initial temperature of the melt, suggest that the paleomagnetic remanence archives up to several thousand to tens of thousands of years succeeding the impact event (i.e., over similar time scales as expected for geodynamo activity resulting in excursions or reversals). Here we report the paleomagnetic results from impact melt and crystalline basement rocks collected from a wide spatial area within the crater, as well as those obtained from three drill cores penetrating the impact melt body. We also present the results of a wide range of rock magnetic experiments (Curie temperature, anisotropy of magnetic remanence, etc.) and paleointensity data. All experiments were performed in laboratories at Ludwig-Maximilians-Universität (Munich).

2. Geological Setting and Sampling

The Manicouagan structure in Québec (Canada, 51.4°N, 68.7°W; Figure 1) formed when a 4–6 km diameter projectile impacted Proterozoic crystalline bedrock at hypervelocity. Several isotopic methods define a Late Triassic age for the impact. The K-Ar method applied to the impact melt yielded 210 ± 8 (2σ) Ma [*Wolfe*, 1971], which was recalculated at 214 ± 8 Ma using revised decay constants [*Steiger and Jäger*, 1978; *Grieve*, 2006]. *Jahn et al*. [1978] obtained an Rb-Sr isochron age of 214 ± 5 Ma, while U-Pb dating on zircons defined a concordant age of 214 ± 1 Ma [*Hodych and Dunning*, 1992]. An initial strontium $^{87}Sr/^{86}Sr$ value of 0.7099 supports the interpretation that the impact melts were derived from the Proterozoic basement [*Jahn et al*., 1978; *O'Connell-Cooper et al*., 2012]. Zircons extracted from the melt sheet yielded 213.2 ± 5.4 Ma based on (U-Th)/He [*van Soest et al*., 2011]; (U-Th)/He applied to titanite separated from anorthosite in the central uplift yielded 208.9 ± 4.1 Ma [*Biren et al*., 2014]. Diaplectic glass from basement rocks lying 10–12 km from the crater's center is suggestive of shock pressures ~35 GPa [*Grieve and Head*, 1983], while stishovite and maskelynite in the central uplift connote peak shock pressures from 12 to 30 GPa [*Biren and Spray*, 2011].

The Manicouagan structure is one of the best preserved and well-exposed impact craters on Earth [*Spray et al*., 2010]. Impact melt sheet occurrences are restricted to the central 55 km diameter island (Île René Levasseur), with rim collapse features occurring on the mainland at ~65 km diameter and beyond. The central island is separated from the mainland by the Manicouagan reservoir, which is part of a hydroelectric program initiated in the 1960s. From 1994 through to 2006, a number of drilling operations took place on the island to appraise the structure as a potential analogue for the Sudbury impact structure of Ontario—a crater renowned for its world-class Cu-Ni sulfide and platinum group element deposits. While no significant economic indicators have been found to date, the drilling programs furnished ~18 km of core, 10 km of which is now held by the Planetary and Space Science Centre at the University of New Brunswick. Moreover, four holes exceed 1.5 km in depth, thus providing an unprecedented window into the deeper structure of the crater stratigraphy [*Spray and Thompson*, 2008]. Specifically, the 1.5 km deep M0608 drill hole, situated close to the geometric center of the structure, penetrated >1 km of clast-free, differentiated impact melt and a further 400 m of clast-laden impact melt before encountering footwall basement rocks. Thicker (~500 m) sections of clast-free impact melt were also present in two other holes, drilled to the north and south of M0608. Other centrally located drill holes to the east and west, with only a thin layer of impact melt, or none at all, aid in delineating a probable fault controlled N-S trough of impact melt and demonstrate a locally castellated structure to the crater floor at Manicouagan [*Spray and Thompson*, 2008]. Previous estimates, based on surface mapping and sampling, placed the melt sheet thickness at 200–300 m [*Floran et al*., 1978].

In August 2010, we drilled and oriented 198 cores at 25 sites in impact melts and basement rocks throughout the crater; most basement rocks come from near the central uplift (Figure 1b). Cores were oriented either manually with a magnetic compass or with an automated system [*Wack*, 2012]. The average declination anomaly from 78 Sun compass readings is −19.8° ± 2.5°, which coincides well with that expected from the international geomagnetic reference field (−20.2°). We also studied the impact melts from three drill holes (M0302, M0603, and M0608; Figure 1b) that penetrated 702 m, 322 m, and 1529 m below the surface, respectively. These deep cores stem from a commercially driven drilling program by three mining companies at 38 locations in the crater. The text below reports depths relative to sea level. Further details surrounding these cores are reported in *Spray and Thompson* [2008], *Spray et al*. [2010], and *O'Connell-Cooper and Spray* [2011].

3. Experimental Results

3.1. Curie Temperature

We measured the temperature dependence of the magnetization with a Petersen Instruments variable field translation balance in an applied field of 30 mT (surface cores) or 37 mT (deep drill cores) (Figure 2). Most impact melt and melt-bearing breccia samples collected near the surface exhibit one or two deflections in the temperature ranges between 530 and 570°C and/or between 600 and 650°C, which are indicative of relatively Ti-poor titanomagnetite and titanohematite, respectively (Figures 2a–2d). Anorthosite basement rocks have Curie temperatures similar to the impact melts, lying generally in the range from 526 to 565°C and around 610°C (Figure 2e). Gneissose diorite from site 25 has Curie temperatures either close to that of stoichiometric magnetite (575°C) or higher (>600°C) (Figure 2f). One observes a high degree of reversibility

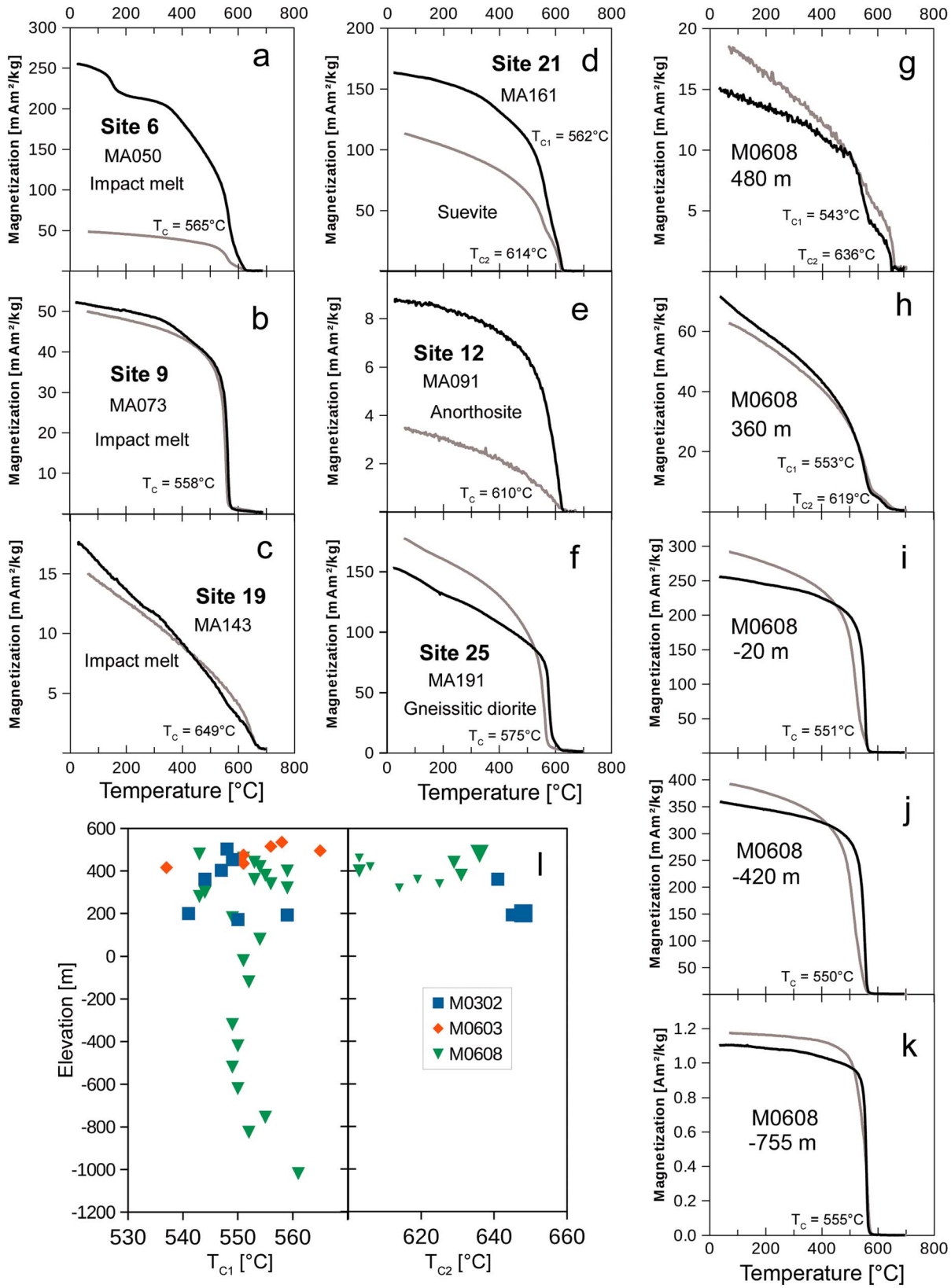


Figure 2. Thermomagnetic curves of samples collected (a–f) on the surface and (g–k) from drill core M0608 as a function of depth. Heating curves are shown in black, cooling curves in gray. (l) Low (T_{C1}) and high (T_{C2}) Curie temperatures from the three drill cores as a function of depth relative to sea level. In M0608, minerals carrying T_{C2} are only found relatively close to the surface. The size of the data points indicates the relative proportion of the magnetization for each of the two populations.

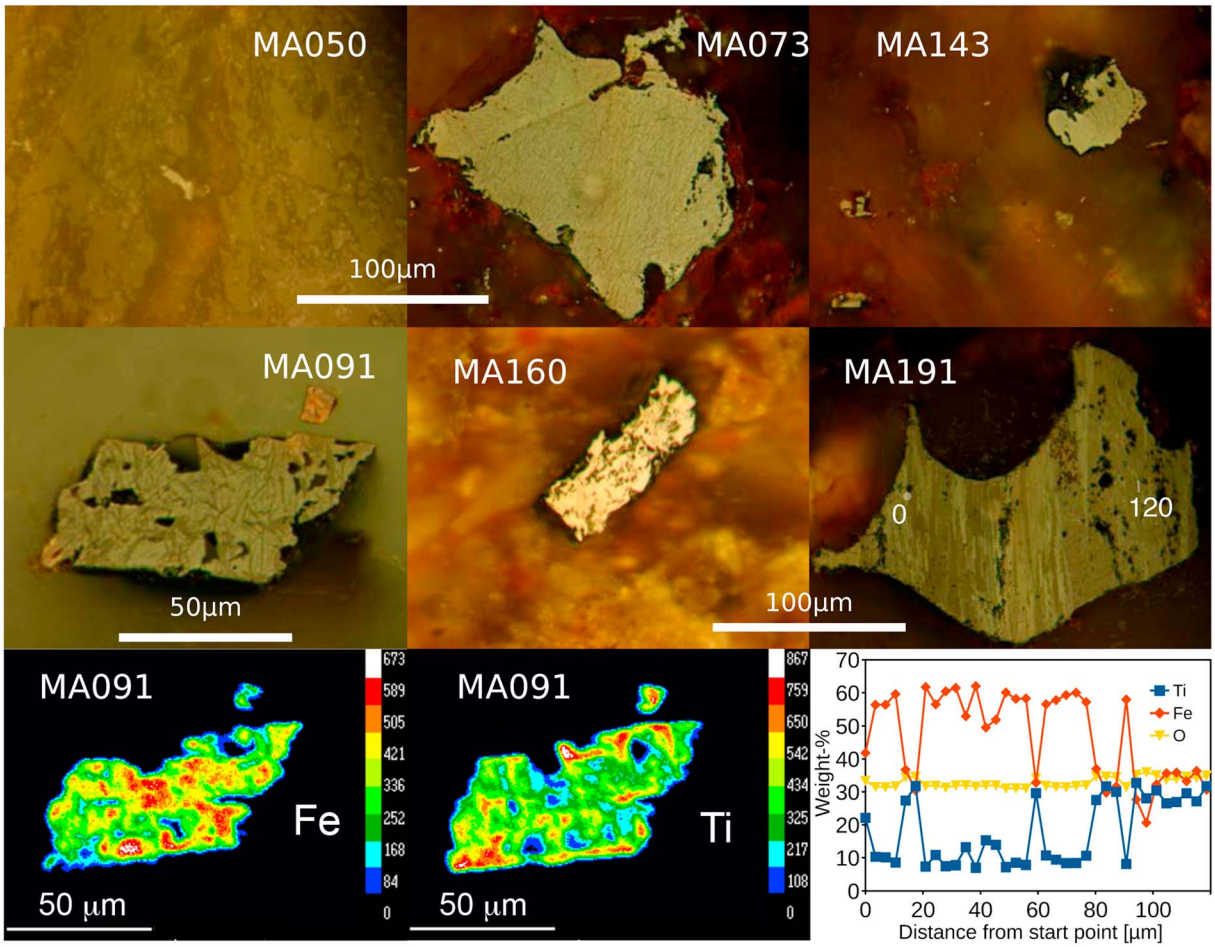


Figure 3. (top and middle rows) Reflected light microscopy images of iron oxide grains from impact melt samples MA050 (site 6), MA073 (site 9), and MA143 (site 13); anorthosite MA091 (site 12); melt-bearing breccia MA160 (site 21); and gneissic diorite MA191 (site 25). (bottom left and middle) Relative Fe and Ti concentrations of an opaque grain from sample MA091 shown in the microscope image above (electron microprobe units in counts per second). Ti-rich, Fe-poor areas correspond to darker phases of the grain. (bottom right) Line profile from electron microscopy of the exsolved grain from sample MA191; start and end points are marked by a circle and a bar in the microscope image; numbers are distances from start point in micrometers. Note that the highly rounded grain morphology in MA191 is also typical of the basement rocks from the Vredefort, South Africa crater (S. Gilder, unpublished data).

between the heating (black) and cooling (gray) curves, meaning that heating in air produces insignificant oxi-exsolution or chemical alteration. In core M0608, the relative contribution of titanohematite in the impact melt samples disappears with depth to where only a single magnetic phase exists with a Curie temperature in the 545–570°C range (Figures 2g–2l). As shown below, samples containing both magnetic mineral phases possess identical magnetization directions, which indicates that both formed during primary cooling in the Late Triassic.

3.2. Reflective Light Microscopy and Electron Microprobe Analyses

Figures 3 (top row) and 3 (middle row) show reflective light images of iron oxide grains from surface samples. Opaque phases rotated in polarized light go extinct, typical of optically isotropic minerals like magnetite. Opaque grain sizes in the impact melts vary by orders of magnitude and show no evidence for shock damage (Figure 3, top row and MA160)—the same observation holds for the basement rocks (Figure 3, middle row MA091 and MA191).

Electron microprobe (Cameca SX100) analyses of all samples reveal exsolved titanium-iron oxides with Ti-rich and Ti-poor zones. Fe and Ti concentration mapping was carried out on an anorthosite sample collected from the central uplift (Mont de Babel) (MA091, Figure 3, bottom left). Total Fe contribution (Fe^{3+} versus

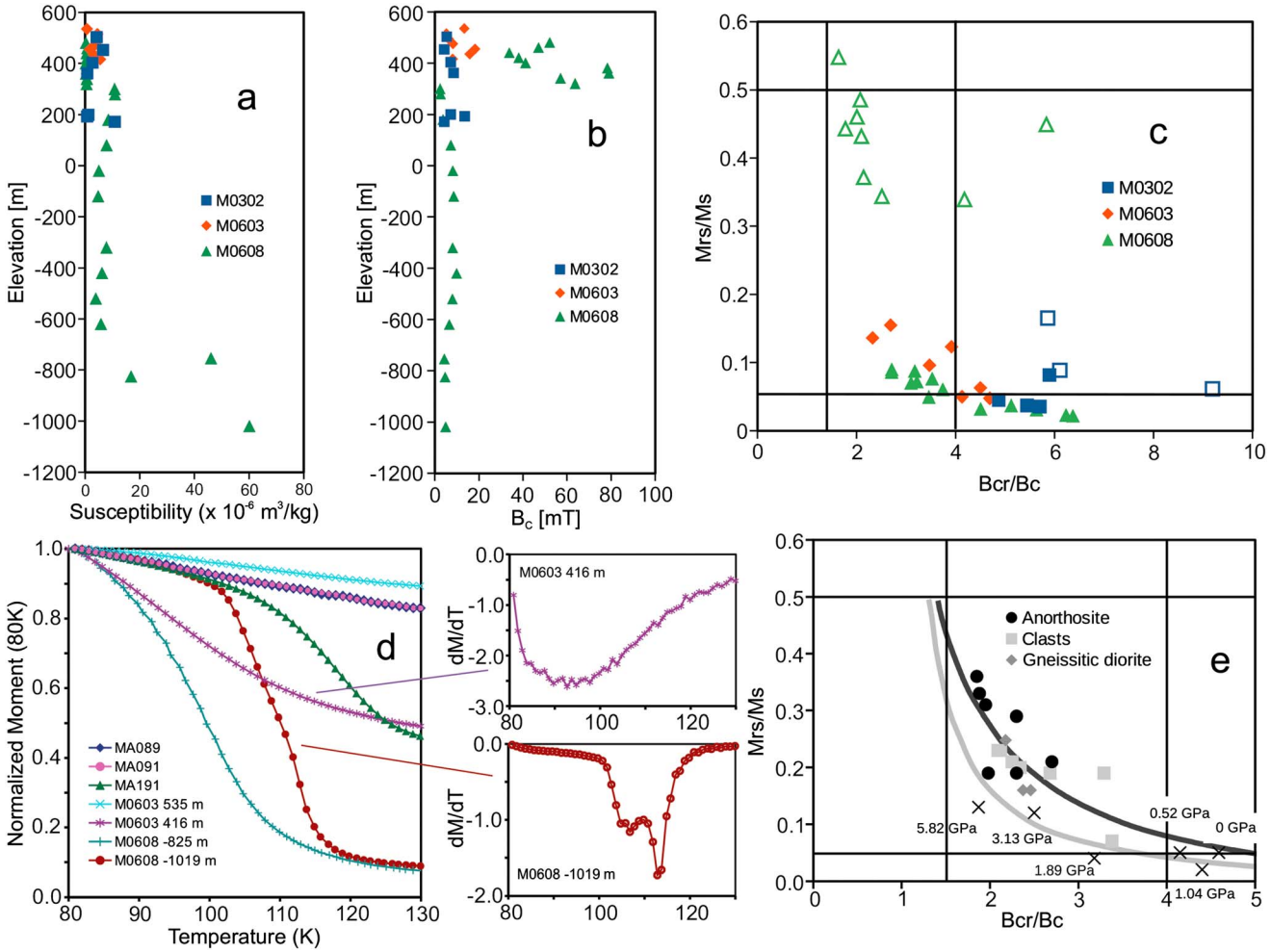


Figure 4. (a) Bulk susceptibility and (b) bulk coercive force (Bc) versus depth with respect to sea level for the samples from the three drill cores. In core M0608, as for the surface cores, samples with the lowest susceptibilities possess Curie temperatures >600°C (titanohematite) as well as higher coercive forces. (c) *Day et al.* [1977] plot of impact melts from the cores. Samples with a significant fraction of Curie temperatures in the 600–650°C range (titanohematite) lie toward the single-domain (SD) field (unfilled triangles), while the rest lie closer to the multidomain (MD) field. Hysteresis loops for the former do not fully saturate in 800 mT fields. Samples from core M0302 that are displaced away from the stoichiometric SD-MD magnetite mixing line [*Dunlop*, 2002] also have a component with Curie temperatures >600°C (titanohematite). (d) Thermoremanence curves of a 1 T field applied at 80 K, then warmed in an ambient field to 130 K to measure variations related to the Verwey transition. Data are normalized to the moment at 80 K. MA089 and MA091 are anorthosites from Mont de Babel; MA191, gneissic diorite (site 25); the rest are impact melts from two cores with depths as indicated. Plots on right are the derivatives of the curves on the left. (e) *Day et al.* [1977] plot of basement rocks from the Manicouagan crater that have only titanomagnetite-like Curie temperatures (<580°C). Anorthosite samples are from Mont de Babel (sites 10 to 12), clasts were entrained in melt-bearing breccia from site 5, and gneissic diorite is from site 25. Crosses denote data points from *Carporzen and Gilder* [2010] for slightly oxidized magnetite measured at ambient conditions after decompression from the indicated pressure. Gray and black mixing lines are from *Dunlop* [2002].

Fe^{2+} in wt %) more closely matches titanohematite than titanomagnetite, which coincides with a Curie temperature at 610°C (Figure 2e). Gneiss sample MA191 (site 25) contains grains with optically visible (tens to hundreds of micrometers wide) lamellae where Ti concentrations average either ~8 wt % or ~30 wt % (Figure 3, bottom right). Fe concentration varies inversely with Ti; oxygen is higher in the zones with higher Ti. The bulk sample has a well-defined Curie temperature of 575°C (Figure 2f), suggestive of nearly pure stoichiometric magnetite. As discussed in the following section, a deflection at 118 K (Figure 4d) matches the Verwey transition of a nearly stoichiometric, yet slightly oxidized, magnetite compatible with a Curie temperature of 575°C. Thus, both titanohematite and titanomagnetite likely coexist in the gneissose diorite.

3.3. Bulk Susceptibility, Magnetic Hysteresis, and Low-Temperature Characteristics

We measured bulk magnetic susceptibility with a Bartington Instruments MS2B sensor. The most striking data are found in core M0608, where susceptibility increases with depth in a few discrete steps (Figure 4a and

Table 1. Site Mean Paleomagnetic Data for Manicouagan[a]

Site	Rock Type	Slat (°N)	Slon (°W)	Alt (m)	*N*/*n*	Dec (deg)	Inc (deg)	*k*	α_{95}	NRM (Am^2/kg)	X (m^3/kg)
1	Impact melt	51.545	68.941	653	7/8	356.1	39.1	147.6	5.0	$1.3 \pm 0.6 \times 10^{-4}$	$6.8 \pm 1.3 \times 10^{-7}$
2	Impact melt	51.547	68.939	608	6/6	356.3	42.0	316.1	3.8	$9.6 \pm 0.3 \times 10^{-5}$	$7.7 \pm 0.4 \times 10^{-7}$
3	Impact melt	51.547	68.939	586	6/6	7.5	43.9	389.3	3.4	$8.3 \pm 0.3 \times 10^{-5}$	$1.5 \pm 0.1 \times 10^{-6}$
4	Suevite	51.318	68.428	359	11/11	15.4	45.0	182.4	3.4	$4.4 \pm 2.5 \times 10^{-4}$	$1.2 \pm 0.4 \times 10^{-6}$
5	Impact melt/clasts	51.318	68.423	359	14/14	13.7	47.7	96.8	4.1	$0.9 \pm 1.4 \times 10^{-3}$	$1.3 \pm 0.6 \times 10^{-6}$
6	Impact melt	51.319	68.426	359	7/7	6.1	49.2	72.6	7.1	$1.5 \pm 0.5 \times 10^{-4}$	$3.0 \pm 1.2 \times 10^{-6}$
7	Impact melt	51.314	68.337	361	6/6	17.1	31.4	202.9	4.7	$8.9 \pm 1.9 \times 10^{-5}$	$1.9 \pm 1.5 \times 10^{-6}$
8	Impact melt	51.313	68.337	384	6/6	19.6	34.7	393.9	3.4	$5.8 \pm 2.3 \times 10^{-5}$	$7.9 \pm 0.9 \times 10^{-7}$
9	Impact melt	51.313	68.337	394	6/6	12.4	38.3	80.1	7.5	$8.1 \pm 4.1 \times 10^{-5}$	$7.0 \pm 5.3 \times 10^{-7}$
10	Anorthosite	51.417	68.699	844	5/6	11.3	39.0	104.5	7.5	$2.4 \pm 3.0 \times 10^{-4}$	$1.9 \pm 0.3 \times 10^{-7}$
11	Anorthosite	51.420	68.696	898	6/6	21.2	41.8	104.3	6.6	$1.9 \pm 1.1 \times 10^{-4}$	$4.5 \pm 2.0 \times 10^{-7}$
12	Anorthosite	51.424	68.692	947	6/6	19.5	38.8	416.1	3.3	$1.3 \pm 0.7 \times 10^{-4}$	$5.6 \pm 3.9 \times 10^{-7}$
13	Anorthosite	51.419	68.596	939	3/6	25.6	39.0	72.6	14.6	$5.5 \pm 4.4 \times 10^{-6}$	$6.3 \pm 3.7 \times 10^{-8}$
14	Anorthosite	51.420	68.595	945	5/6	20.6	43.5	226.7	5.1	$1.8 \pm 1.1 \times 10^{-6}$	$2.7 \pm 1.0 \times 10^{-8}$
15	Impact melt	51.395	68.846	510	6/6	13.2	34.6	382.4	3.4	$3.0 \pm 1.3 \times 10^{-4}$	$6.8 \pm 9.4 \times 10^{-7}$
16	Impact melt	51.215	68.856	470	6/6	16.3	38.7	104.5	6.6	$4.4 \pm 0.8 \times 10^{-5}$	$9.0 \pm 4.8 \times 10^{-7}$
17	Impact melt	51.232	68.577	360	6/6	15.9	39.9	202.9	4.7	$8.2 \pm 3.9 \times 10^{-5}$	$2.0 \pm 0.6 \times 10^{-6}$
18	Impact melt	51.235	68.578	361	6/6	30.5	39.7	923.4	2.2	$4.0 \pm 0.2 \times 10^{-4}$	$6.4 \pm 0.2 \times 10^{-7}$
19	Impact melt	51.235	68.578	356	6/6	18.6	41.8	757.1	2.4	$1.7 \pm 0.5 \times 10^{-4}$	$4.6 \pm 1.8 \times 10^{-7}$
20	Impact melt	51.490	68.338	360	6/6	13.7	36.6	168.2	5.2	$3.1 \pm 0.6 \times 10^{-4}$	$6.1 \pm 1.2 \times 10^{-6}$
21	Suevite	51.490	68.339	353	6/6	6.5	46.6	180.5	5.0	$8.7 \pm 1.4 \times 10^{-4}$	$4.4 \pm 0.5 \times 10^{-6}$
22	Impact melt	51.489	68.339	352	6/6	11.4	44.0	145.1	5.6	$1.1 \pm 0.9 \times 10^{-3}$	$1.2 \pm 0.5 \times 10^{-5}$
23	Impact melt	51.507	68.344	355	6/6	21.3	38.3	277.6	4.0	$2.5 \pm 0.5 \times 10^{-4}$	$3.0 \pm 1.3 \times 10^{-6}$
24	Impact melt	51.507	68.344	363	6/6	11.6	39.1	252.3	4.2	$2.4 \pm 0.5 \times 10^{-4}$	$8.1 \pm 3.0 \times 10^{-6}$
25	Gneissic diorite	51.507	68.345	361	5/6	9.6	47.8	682.4	2.9	$6.8 \pm 2.0 \times 10^{-4}$	$3.2 \pm 0.5 \times 10^{-6}$

[a]Slat, site latitude in degrees north; Slon, site longitude in degrees west; Alt, altitude in meters; *N*/*n*, number of samples used in the calculation/number of samples stepwise demagnetized; Dec, declination; Inc, inclination; *k*, best estimate of the precision parameter; α_{95}, the radius of the cone where the true mean direction lies within 95% confidence; NRM, average and single standard deviation of the mass normalized, natural remanent magnetization; χ, average and single standard deviation of the mass normalized susceptibility.

Table S1 in the supporting information). Above 300 m, the mass normalized susceptibility averages 0.5×10^{-6} m^3/kg, then 7×10^{-6} m^3/kg from 300 m to −620 m, and then jumps to $20–60 \times 10^{-6}$ m^3/kg below −620 m. Saturation magnetization obtained from hysteresis loops (Petersen Instruments, variable field translation balance) mimic the pattern displayed by susceptibility (Table S1). As the mass susceptibility of titanomagnetite is orders of magnitude greater than titanohematite and paramagnetic, iron-bearing minerals like pyroxene [*Dunlop and Özdemir*, 1997], one can use susceptibility as a proxy for titanomagnetite concentration.

Bulk coercive force (Bc) defines two discrete populations in M0608 with average values of 54 mT above 320 m and 6 mT below (Figure 4b). Together with the Curie temperature data, this behavior is attributed to a profusion of titanohematite in the upper part of the impact melt sheet. Below, titanomagnetite is present as the only carrier of the magnetization with an increasing abundance toward the base. As expected, titanohematite-bearing samples plot closer to the single-domain field on a Day plot (Figure 4c) [*Day et al.*, 1977], with the caveat that not all samples above 320 m were fully saturated by 800 mT. The rest do reach saturation and lie closer to the multidomain field. For this population, titanohematite-bearing samples depart from the pure-stoichiometric magnetite mixing line of *Dunlop* [2002]. These observations suggest that oxidation was prevalent above 320 m, where titanohematite is readily observed in the surface samples (Tables 1 and S1). However, oxidation at the surface is heterogeneous, as some sites (3, 22, and 24) contain no, or minimal traces of, titanohematite. We found no spatial correlation for the occurrence of titanohematite. Because the solely titanomagnetite-bearing impact melts lie near or within the multidomain field, erosion must have removed the uppermost layers that cooled fast enough to produce smaller, single-domain-sized grains.

The prevalence of oxidation toward the surface can also be seen in the thermomagnetic curves measured with a Lakeshore MicroMag 3900 vibrating sample magnetometer (Figure 4d). The samples were cooled in the ambient field to 80 K, exposed to a 1 T field at 80 K, and then warmed from 80 to 130 K in the ambient field. For core M0603, the sample lying closest to the surface at 535 m has a Curie temperature of 558°C (Table S1), which corresponds to a Ti concentration of 0.9 wt % [*Akimoto*, 1962]; thus, it should possess a visible Verwey transition. Oxidation likely explains the absence. The sample at 416 m has a broad Curie

temperature averaging 537°C, which corresponds to an average Ti concentration of 1.8 wt %. A diffuse Verwey transition at ~93 K is consistent with the Ti concentration [*Kakol et al*., 1994]. However, the drop in moment in the 80–130 K window is not as substantial as one would expect if the sample possessed a high fraction of multidomain grains, as expected from the hysteresis parameters (Table S1), and that over 95% of the remanence was demagnetized by an alternating field with a peak field of 75 mT (see below, Figure 7d). Thus, the titanomagnetite in this sample is also likely oxidized, albeit not enough to completely erase the expression of the Verwey transition as in the sample 119 m above.

Samples from the deepest levels of M0608 at −825 and −1019 m also show Verwey transitions typical of multidomain grains owing to the significant drop in magnetization [*Moskowitz et al*., 1993]. The Curie temperatures of these samples are 552 and 561°C, respectively (Ti concentrations of 1.2 and 0.8 wt %), so we should expect a lower Verwey transition temperature for the sample at −826 m than at −1019 m, which is consistent with the data (Figure 4d). Interestingly, the deepest sample shows two distinct low-temperature transitions yet only a single Curie temperature in the titanomagnetite range (Table S1).

The sum of these observations is entirely compatible with Mössbauer spectra on Manicouagan impact melts, which show that hematite dominates the mineral phases with the highest ferric iron concentrations ([*Morris et al*., 1995; note that this study only considered end-member Fe compositions, i.e., no Ti]). *Morris et al*. [1995] likewise found that hematite and magnetite can coexist in the same rock in widely varying proportions; the lack of iron hydroxide or maghemite led them to conclude that the oxidation occurred above 250°C. We also find no evidence for either iron hydroxide or maghemite. *O'Connell-Cooper and Spray* [2011] found higher ferric oxide (hematite) concentrations toward the top of core M0608. This coincides with our findings that titanohematite, likely caused by high-temperature oxidization, disappears by a depth of 320 m in core M0608. That titanomagnetite increases with depth is consistent with their finding that fractional crystallization occurred in the lower portion of the core. Moreover, the melt sheet must have cooled slowly enough to create multidomain-sized grains. On the other hand, *O'Connell-Cooper and Spray* [2011] report a decrease in magnetite with depth, which is at odds with the susceptibility and hysteresis (saturation magnetization) data. We find only the presence of titanomagnetite in core M0603, consistent with *O'Connell-Cooper and Spray* [2011], who detected no appreciable ferric iron component. It is noteworthy that the basement rocks and impact melts have similar magnetic mineralogies.

Figure 4e shows a Day plot of the basement rocks (including basement clasts entrained in melt-bearing breccia or in impact melt at the base of the impact melt sheet) for samples that only have a dominant Curie temperature below 580°C. Titanomagnetite-bearing, deep-seated plutonic rocks like anorthosite and diorite would typically plot in the multidomain field on this graph, yet the samples from Manicouagan lie well within the pseudo single-domain field. There are three potential explanations. One unlikely scenario is that titanohematite, clearly present in many surface samples, is not expressed in the Curie temperature measurements but influences the remanence and coercivity ratios. Although minor amounts of titanohematite could alter the coercivity ratio, it should not raise the remanence ratio. Second is that high-temperature oxidation reduced the grain size fraction, displacing preimpact titanomagnetite grains that would originally lie in the multidomain field toward the single-domain field on the Day plot. Such an effect was observed at the Mistastin crater, where baked anorthosite within 1 m of the contact with impact melt has higher coercivity than nonbaked anorthosite lying farther away from the contact [*Hervé et al*., 2015, Figure 6]. Third, pressure cycling of slightly oxidized multidomain magnetite makes it more single-domain-like [*Gattacceca et al*., 2007; *Carporzen and Gilder*, 2010]. Figure 4e superimposes the pressure-cycled data on the Day plot, which shows the evolution that multidomain magnetite undergoes when pressurized up to 5.8 GPa. As the pressures involved at Manicouagan exceed 30 GPa, it is not surprising that some samples plot well above the experimental result at 5.8 GPa. Moreover, *Hervé et al*. [2015] found that the anorthosite basement rocks at Mistastin trend from multidomain to single domain when approaching the center of the crater where pressures were greater. In sum, we interpret that the higher than expected remanence ratios and lower than expected coercivity ratios of the titanomagnetite-bearing basement rocks at Manicouagan reflect both pressure and temperature effects related to the impact.

3.4. Stepwise Demagnetization and Paleomagnetic Directions

All samples were stepwise demagnetized using either 10 steps with a peak alternating field up to 75 mT or 11–15 steps using thermal demagnetization up to 680°C. Remanent magnetization measurements were performed in magnetically shielded rooms using 2G Enterprises Inc., three-axis superconducting quantum

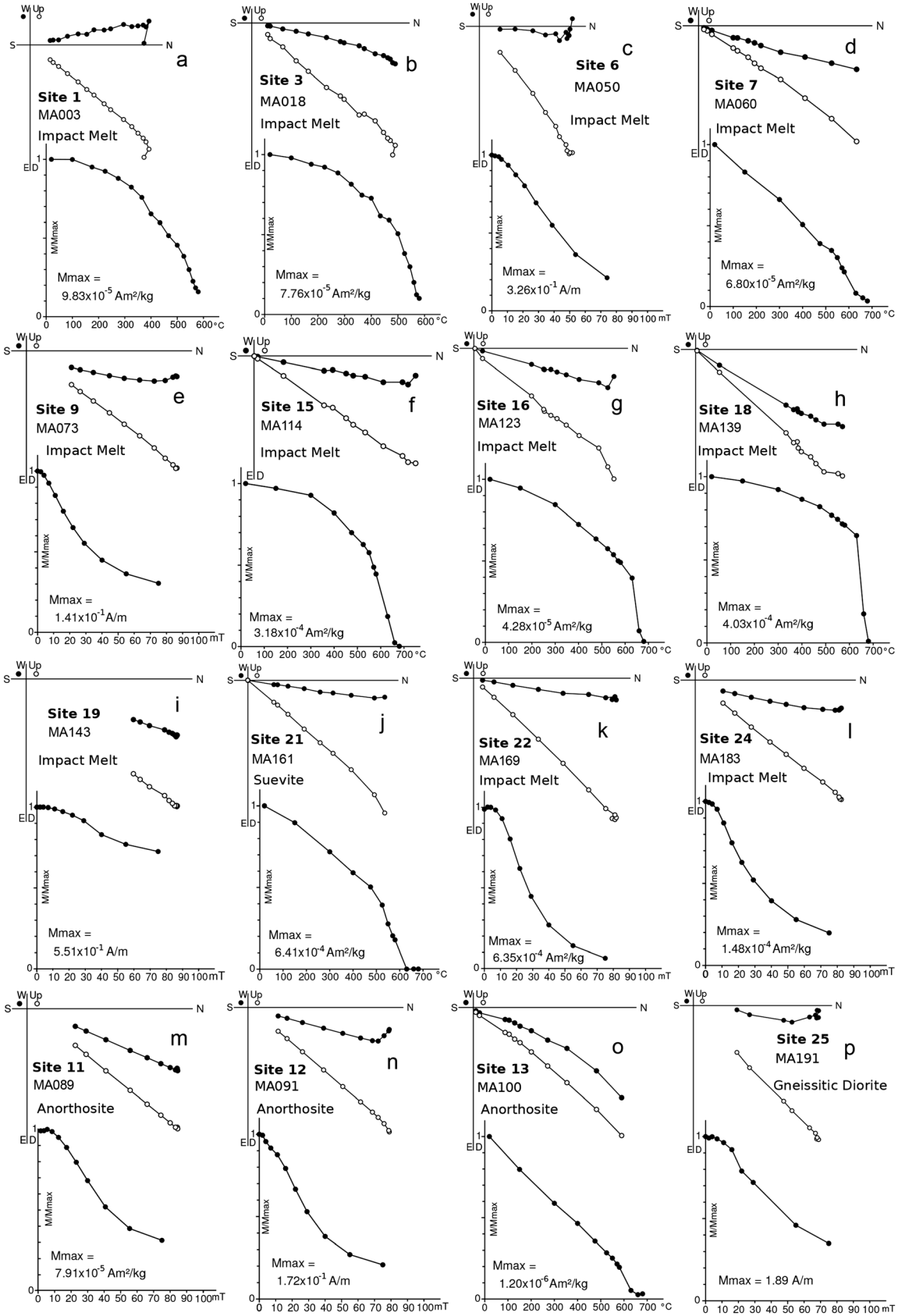


Figure 5. Orthogonal projections of stepwise demagnetization and normalized magnetization decay plots for samples collected on the surface.

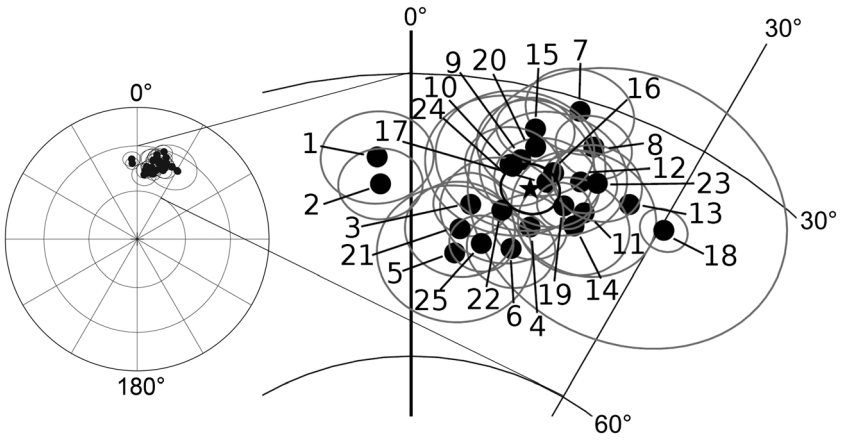


Figure 6. Stereographic projection of the site mean directions with α_{95} in gray (data from Table 1). The star with black α_{95} denotes the mean of the site means.

interference device magnetometers, either manually using the Cryomag program [*Wack*, 2010] or with the SushiBar automated system [*Wack and Gilder*, 2012]. Remanent magnetization directions were determined with principal component analysis [*Kirschvink*, 1980]; Fisher statistics [*Fisher*, 1953] were applied to the site mean and overall mean directions.

Demagnetization characteristics of the impact melts (Figures 5a–5l) are similar to those of the basement rocks (Figures 5m–5p). Both thermal and alternating field demagnetization unblock magnetization components that decay univectorially to the origin after the first few demagnetization steps. Despite the presence of a single magnetization direction, normalized thermal demagnetization decay plots usually identify multiple inflections at temperatures consistent with the in-field thermoremanent data (Figure 2), e.g., indicative of titanomagnetite and titanohematite. Alternating field demagnetization to 75 mT removes nearly all of the original moment in some samples (Figure 5k), while for others it is less effective (30% in Figure 5i, 70% in Figure 5e). Magnetization directions for the latter case still trend toward the origin, which shows that the higher coercivity component possesses the same direction as that with lower coercivity.

An important aspect of the demagnetization data is that a single magnetization direction of normal polarity was recorded by all magnetic phases in each sample, meaning that the blocking of the remanence was coeval for all magnetic recorders in a given rock. This observation holds true even when accounting for directional differences among samples throughout the crater, which vary enough to be visible (if present) on the orthogonal plots. The fact that we observe no such difference implies that the creation of titanohematite (oxidation) occurred above titanomagnetite Curie temperatures and that no subsequent alteration by weathering or hydrothermal activity influenced the ferrimagnets. Hence, the titanohematite likely originated from high-temperature oxidation that occurred during degassing and not from convection or diffusion that brought oxygen down into the melt from the surface.

Sample directions within sites are generally well grouped, with the best estimates of the precision parameter (k) $>$100 in 21 of 25 sites (Table 1). Only site 13 (anorthosite from Maskelynite Peak) has an α_{95} (the radius of the cone about which the true mean lies with 95% confidence limits) exceeding 10°, although its mean is based on only three samples with relatively weak magnetizations (Table 1). Site mean directions span 35° in declination and 18° in inclination resembling a range typical of that observed over the last few millennia (Figure 6) [*Bucur*, 1994; *Hervé et al*., 2013]. Sites 1 and 2 have the most westerly declinations, yet they were sampled nearby site 3 whose mean direction is distinct from them at 95% limits. We observed no fault between these three sites, whose elevations (600–650 m; Table 1) are the highest of any impact melt sampled in this study. Sites 17–19 were also sampled in close proximity, yet the direction of site 18 lies well away from the other two, at the easternmost extremity of all site mean directions. Interestingly, site 18 is an impact melt dike, so its magmatic history could be different from the nearby sites if it intruded and cooled at a later time.

Of importance is that the site mean directions from the basement rocks lie well within the distribution of the impact melts. The average direction of the impact melts and of the basement rocks is indistinguishable at

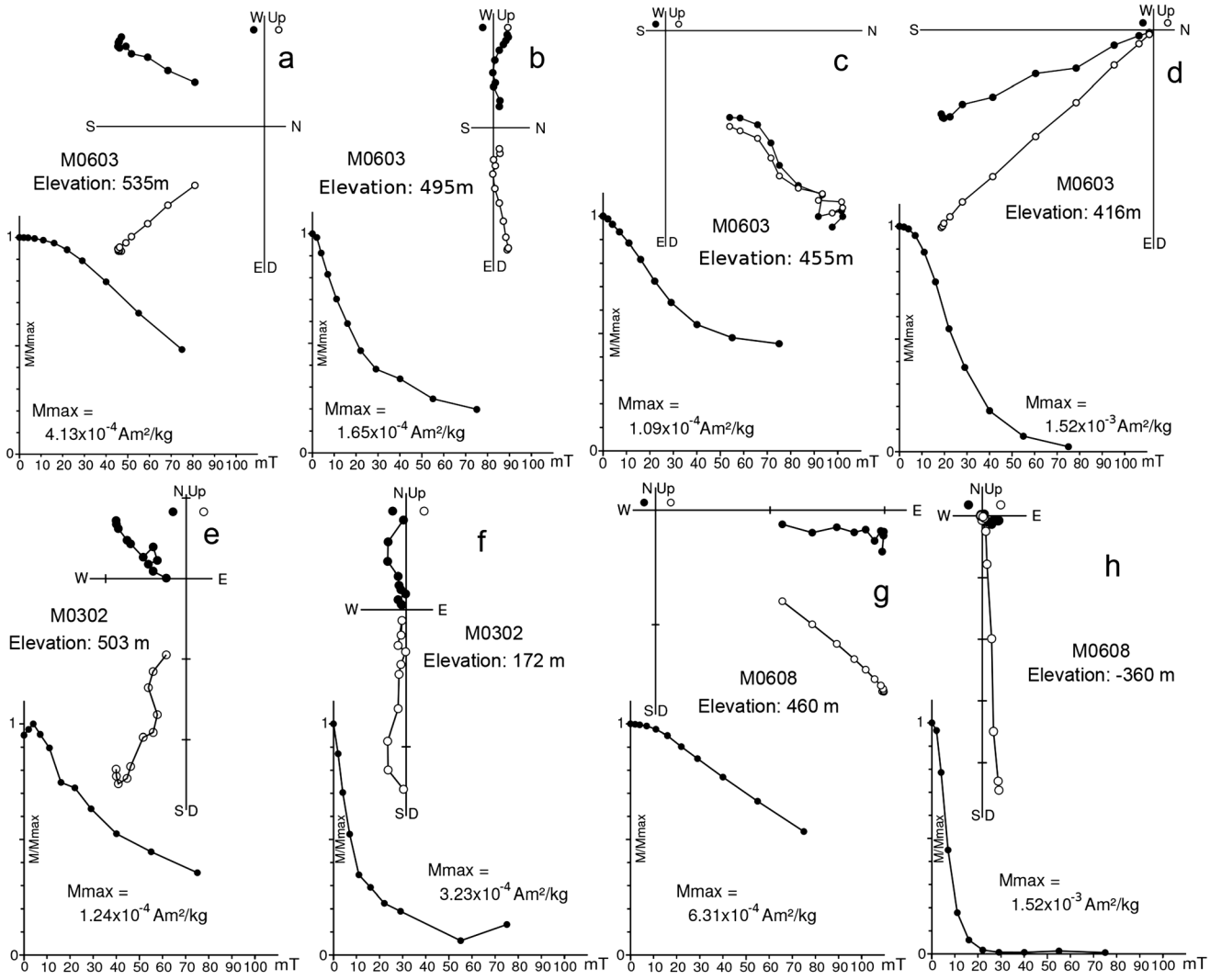


Figure 7. Orthogonal projections of stepwise demagnetization and normalized magnetization decay plots for samples from the deep drill cores.

95% confidence limits. This suggests that the basement rocks acquired their remanence directions in the Triassic, likely from thermal overprinting. The overall mean direction of our 25 sites (declination [D] = 14.2°, inclination [I] = 41.1°, k = 125, α_{95} = 2.7°) is indistinguishable within 95% confidence limits from that of *Larochelle and Currie* [1967] (D = 12.0°, I = 41.2°, k = 91, α_{95} = 4.8°, N = 11 sites) based on directions derived from a single demagnetization step at 20 mT (no principal component analysis) and excluding one site (their site 11). The overall average paleomagnetic pole from our study is longitude = 84.7°E, latitude = 60.2°N, and A_{95} = 2.8°).

Figure 7 shows stepwise alternating field demagnetization results of the impact melts from the deep drill cores, which were drilled vertically, except for M0302 that was drilled 60° from horizontal. The declination component is arbitrary. The room temperature (natural) remanent magnetization inclinations separate into two general bands: relatively shallow (20° to 45°) and relatively steep (60° to 90°) (Figure 8a). Demagnetization trajectories in most samples from core M0603 decay toward the origin and yield inclinations based on line-fit analysis between 35.8° and 48.8° with an average of 40.0 ± 5.4° (Figures 7a–7d and 8b), within 95% confidence limits of 40.8 ± 2.2° calculated from inclination-only statistics on the site mean data (Table 1). Neither a systematic change in inclination nor any significant deviation in inclination occurred within the time represented by the 119 m of impact melt that cooled through the titanomagnetite blocking temperature.

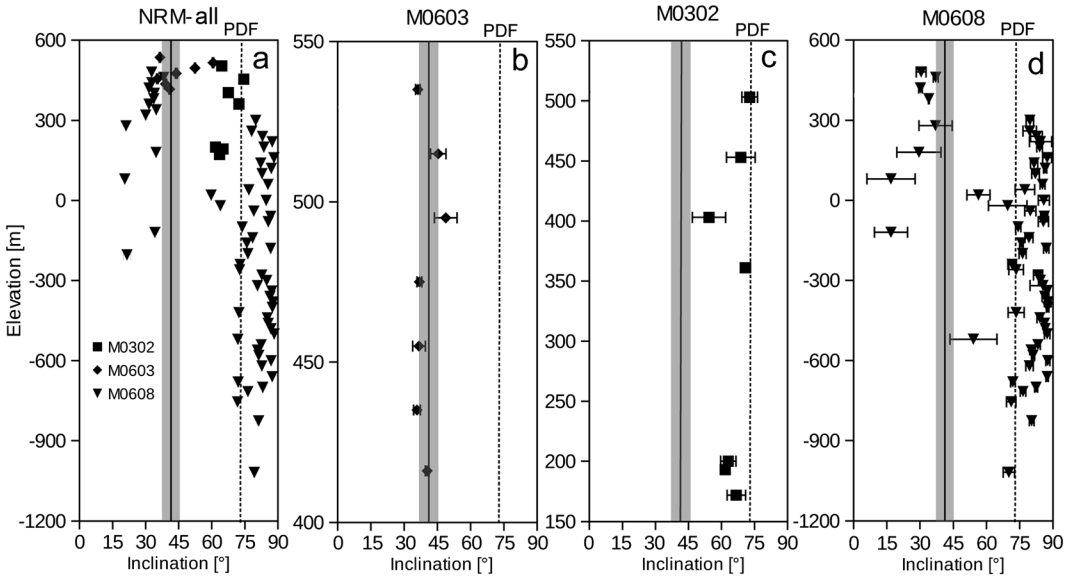


Figure 8. Paleomagnetic inclinations of drill core samples as a function of elevation relative to sea level. Uncertainties are the angular deviations associated with the line fitting procedure used to determine the magnetization directions. The black line at 41.1° represents the overall average inclination with 2σ uncertainty limits in gray (Table 1). The dashed line is the present-day field (PDF) inclination (73.8°); the expected axial centric dipole inclination is 68.2°. (a) Inclinations of the natural remanent magnetization (NRM), which is measured at room temperature. (b–d) Inclinations derived from principal component analysis of the demagnetization spectra.

Demagnetization trajectories from M0302 and M0608 are more complex than in M0603 (Figures 7e–7h). Most samples from M0302 and M0608 possess inclinations 60° or greater (Figures 8c–8d), which we interpret as drilling-induced remagnetizations, since they lie near the expected present-day field inclination (73.8°) at Manicouagan, or parallel to the core's axis (90°). Some scatter in the inclination data in M0302 could be related to the nonvertical drill direction. Although some samples have shallower inclinations, including a trend toward shallow (15°) inclinations in M0608 with depth, we are loath to interpret them in a geomagnetic sense given the propensity of effects related to drilling. Hence, samples collected at the surface, as well as those from M0603, display a pattern typical of secular variation. These deposits span a total thickness from 653 m (site 1) to 172 m (lowest sample in core M0603), meaning that no significant aberration in geomagnetic field direction occurred in the time it took to cool 481 m of impact melt through the blocking temperatures of the ferrimagnets.

3.5. Paleointensity

Absolute paleointensity experiments were performed according to the multispecimen, parallel differential, partial thermal remanent magnetization (pTRM) protocol of *Dekkers and Böhnel* [2006] on 186, 5 mm diameter cylindrical specimens taken from outcrops and from core M0603 that escaped drill-induced overprinting. The multispecimen method was selected in preference to the classic Thellier-Thellier-type experiments given the complex magnetic mineralogies and propensity of multidomain-sized grains in the Manicouagan samples. Moreover, the magnetization of the samples rarely alters when heating in air at high (>500°C) temperatures (Figure 2). We only used specimens if the angle between their inclination measured at room temperature deviated less than 60° from the core's *z* axis, which was the applied field direction in the laboratory. Sample magnetizations were measured after demagnetizing in two steps at 100°C and 500°C in a null field. After null field heating to 500°C, we again heated them to 500°C and cooled them down to room temperature in the presence of a magnetic field (pTRM step), after which the samples were thermally cycled to 100°C in a null field before their magnetizations were measured. A heating threshold of 500°C was chosen because it was low enough to avoid alteration and at least 20% of the natural remanent magnetization decayed by 500°C in stepwise thermally demagnetized samples (Figure 5). The pTRM step was made in six discrete fields from 10 to 60 μT for the surface samples and in eight discrete fields from 10 to 45 μT for the drill core samples.

Figure 9 shows the median normalized (pTRM-TRM)/pTRM values and standard deviations obtained at each field increment after subtracting the magnetization acquired below 100°C to avoid minor overprints. The

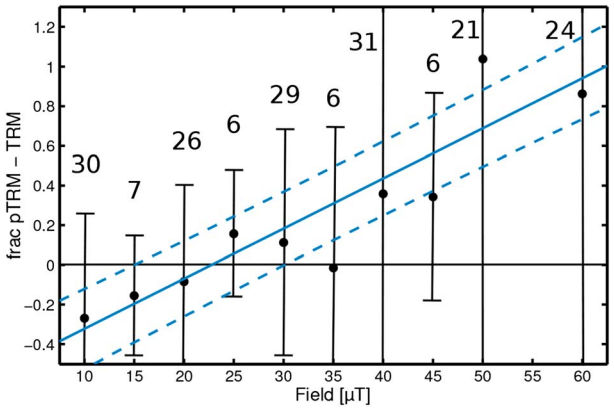


Figure 9. Paleointensity measured with the multispecimen parallel differential pTRM method [*Dekkers and Böhnel*, 2006]. Black circles are the median values; the number of samples used at each field is given together with the single standard deviation. The solid blue and dashed lines represent a linear fit and one standard deviation of the data. The zero-crossing point is at 22.8 ± 7.7 μT.

solid blue line represents the linear fit of the data with the 1σ confidence envelope (dashed blue line). The zero-crossing point of the fit corresponds to the paleointensity (22.8 ± 7.7 μT). Treating the surface and the M0603 core samples separately yields paleointensities of 21.0 ± 11.8 μT and 25.4 ± 14.4 μT, respectively.

3.6. Anisotropy of Magnetic Remanence

The anisotropy of anhysteretic remanent magnetization was measured on 191 samples with a decaying alternating field between 20 and 0 mT and a bias field of 50 μT using the automated SushiBar system with a 12 position protocol [*Wack and Gilder*, 2012]. Samples possessing inherited clasts were excluded, which explains the absence of data from sites 4 and 5. Table S2 lists the mean eigenvalues, the principal axis directions, the corrected anisotropy degree (*P'*) [*Jelinek*, 1981], and the shape parameter (*T*) [*Jelinek*, 1981] for each site. The impact melts have an average anisotropy degree of 13%, while that for the basement rocks is 20%. Most fabrics are triaxial (*T* around 0).

Figure 10a shows stereonet plots of the bootstrapped principal axis directions at the different impact melt sites in the crater. Data from sites collected in close proximity were pooled together. At each location, we calculated the radial angle with respect to the crater's center and plotted it against the declination of the major anisotropy axis (M_1) (Figure 10b). Two solutions are possible for the M_1 axis declinations. Figure 10b displays the option that lies in a band of ±90° with respect to the radial angle, where it is seen that six of seven points deviate less than ±45° from the 1:1 line. In the case of a completely random relationship between radial angle and M_1 declination, the probability for each datum to lie no more than 45° from the 1:1 line is 1/2. Thus, the probability of having six of seven points within ±45° is 5% for a purely random process. If we

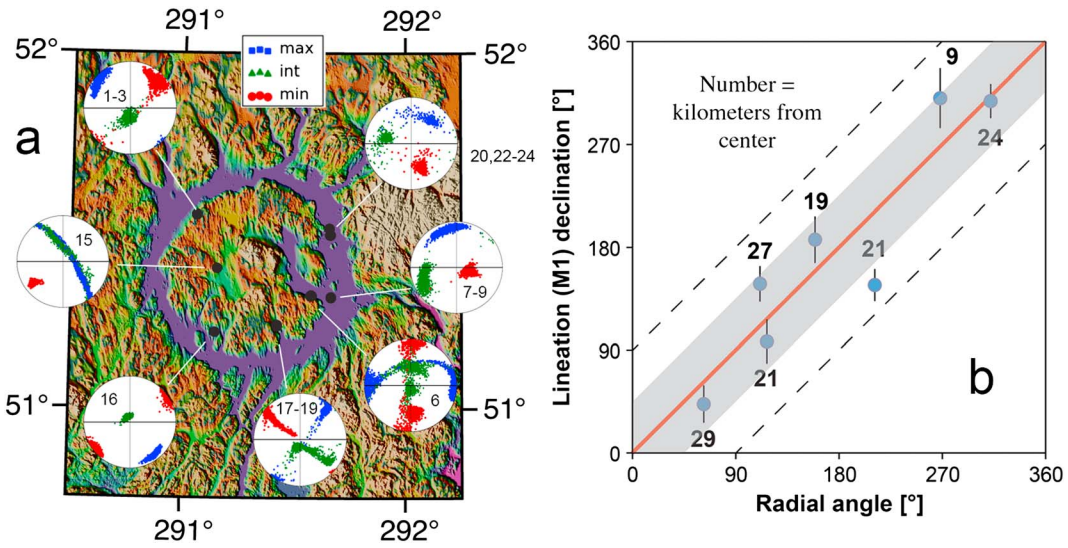


Figure 10. (a) Topographic map of the Manicouagan crater with the bootstrapped anisotropy of remanent magnetization principal axis directions for each group of impact melt sites (black circles). (b) Declination of the major axis (M_1) of anisotropy of magnetic remanence (Table S2) versus radial angle from the crater's center. The 1:1 line is shown in red, the ±45° swath about the 1:1 line is in gray, and the dashed line delimits ±90° swath about the 1:1 line.

count site 15, which lies closest to the crater's center of all the sites, and whose uncertainty does not exclude the case of lying outside the ±45° band, then the probability for randomness becomes 16%.

The idea behind making Figure 10 originated during fieldwork near site 24 where a clast-laden impact melt exhibits a well-developed, ductile flow fabric whose lineation trends radially from the crater's center (Figure 1e). Although no fabric is readily observable by eye in most impact melt locations, Figure 10b implies that there is a general correlation between the trend in the flow lineation versus location in the crater; e.g., the fabrics trend radially away from/toward the crater's center. This may have been caused during crater formation, when the impact melts moved outward (centrifugally) from their site of origin, or during (centripetal) back flow of the impact melt toward the crater's center [*Dence et al*., 1977]. Macroscopic deformation fabrics near site 24 suggest flow was outward.

Such fabrics are normally impossible to resolve in inch (2.5 cm) cores because inherited clasts entrained in clast-bearing impact melt or melt-bearing breccias have their own, preimpact fabrics that obscure any fabric related to flow. The selected Manicouagan impact melts are largely devoid of inherited clasts, leading to a more robust interpretation of the anisotropy data in terms of a primary flow fabric. One could expect that the fabric would be better defined closer to the crater margins where the shear component due to oversteepened walls would be greater, which is why Figure 10b lists distance from the center in kilometers. However, a potential correlation is obscured since most sites were collected near the annular lake between 20 and 30 km, nor can one discount radially directed flow during uplift of the central peak. Given the wide range in fabric type (oblate, triaxial, and prolate), we do not expect all sites to correlate well; further dedicated work on this topic seems warranted.

4. Interpretations

4.1. Crater Formation Processes

Hydrothermal processes in impact craters have received increasing attention due to their potential importance for the origin of life on Earth and other planets [e.g., *Osinski et al*., 2001; *Abramov and Kring*, 2005]. However, the spatial distribution of the iron oxides at Manicouagan is better explained by a model of degassing and oxygen diffusion in the upper few hundred meters of the impact melt, rather than an advecting system associated with deeper-seated hydrothermal activity. If a hydrothermal plumbing system once existed within the uppermost part of the crater stratigraphy, any trace of it has been removed by erosion.

The magnetic anisotropy data provide a unique opportunity to track impact melt flow kinematics present during the earliest stages of impact crater development. *Melosh and Ivanov* [1999] calculated that the transient crater collapses within 100 s after impact. Thus, outward flow or return of the melt as it flowed back into the crater must have occurred relatively soon after impact, and the melt viscosity must have been high enough to produce a systematic alignment of mineral grains. Titanomagnetite and titanohematite crystallize from $>$1000°C to as low as 600–700°C depending on magma chemistry [*Buddington and Lindsley*, 1964; *Hargraves and Petersen*, 1971; *Lipman*, 1971], so if the fabric truly reflects complex crater formation processes, the molten material must have fallen below its liquidus and viscosity increased within minutes after impact. Clearly, a considerable time lag exists between crater formation and when the magnetic remanence was acquired (section 4.3). It would seem that shear should be higher near the contact with the basement floor, which is hard to ascertain. For complex impact craters, a central uplift grows as the floor rebounds [e.g., *Grieve et al*., 1981]. Because the final distribution of the impact melt appears to be controlled by faults surrounding the central uplift (Mont de Babel) [*Spray et al*., 2010], it could be that a growing central uplift caused the impact melt to flow radially away from it. Hence, the radial distribution of the maximum anisotropy axes could be due to outward or back flow into the collapsing transient crater and/or flow away from the emerging central uplift.

4.2. The Central Magnetic Anomaly

A positive magnetic anomaly with an amplitude of 2000 nT lies just southwest of the central uplift (Figure 1b). Whereas *Coles and Clark* [1978] attributed the distinct magnetic anomalies outside the crater to iron ore deposits, they could not correlate the strong central magnetic anomaly with the local geology. High horizontal magnetic field gradients led them to propose that the anomaly's source must be relatively shallow. Their model used a paleomagnetic direction of $D = 10°$ and $I = 40°$ [*Larochelle and Currie*, 1967; *Robertson*, 1967]

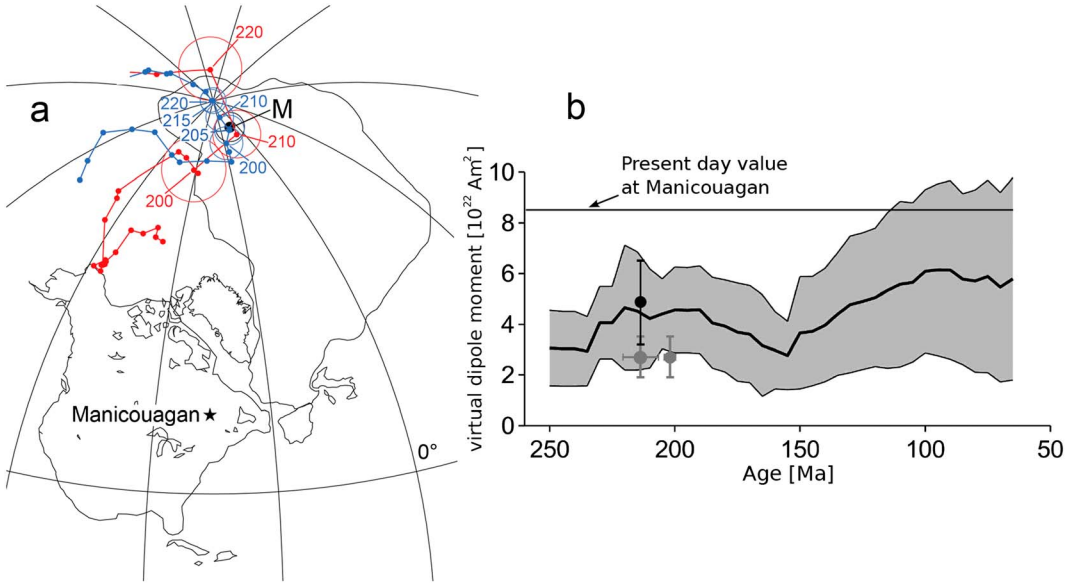


Figure 11. (a) Paleogeographic reconstruction in a 214 Ma reference frame with respect to North America (Euler parameters from *Torsvik et al*. [2001]). Apparent polar wander paths of North America are shown in blue [*Torsvik et al*., 2001] and red [*Besse and Courtillot*, 2002]; numbers are ages in megayears. The paleomagnetic pole for Manicouagan (M, this study) is in black. (b) Evolution of the virtual dipole moment during the Mesozoic using a running average (thick black line) with a window size of 20 Ma; the gray region represents the single standard deviation (PINT database of *Biggin et al*. [2010]). The black circle with black 1σ uncertainty envelope denotes the virtual dipole moment from Manicouagan, and gray circle with gray uncertainty envelope denotes the virtual dipole moment from the Rochechouart crater for 214 ± 8 Ma [*Kelley and Spray*, 1997] and 201 ± 2 Ma [*Schmieder et al*., 2010].

with an annular-shaped source body, 8×12 km, extending 2 km below the surface with a magnetization of 11 A/m. Given our new findings, only the deeper levels of the impact melt with the more enriched titanomagnetite can satisfy such large magnetization values.

Scott et al. [1996] analyzed a drill core that penetrated basement rocks near the eastern part of the anomaly to a depth of 472 m (S in Figure 1b). Relatively high magnetic susceptibility (10^{-3} to 10^{-1} SI) and high remanent magnetizations (0.1 to 100 A/m) led them to ascribe the magnetic anomaly to basement rocks that became enriched in magnetite due to shock decomposition of garnet and pyroxene, followed by hydrothermal alteration that favored iron oxidation. However, if hydrothermal activity and shock related to impact did create new magnetite, we would expect the paleomagnetic inclinations to be consistent with that of the impact (~40°), yet the inclinations from their core are $<30°$, which leads us to suspect the presence of a drilling-induced overprint. On the other hand, mafic gneisses within the central anomaly region have susceptibility values equal to or greater than the deep differentiated impact melts (L. Thompson, unpublished data), so the mafic gneisses likely play a significant role in producing the positive anomaly. Although the deeper impact melts contribute to the central anomaly's structure and amplitude, their influence appears much less than that of the underlying mafic gneisses.

4.3. Effect of Impact on the Geodynamo

Stepwise demagnetization reveals that all rocks from the Manicouagan impact structure possess a single normal magnetization component whose overall mean likely averages out geomagnetic secular variation. Converting the overall mean direction into a virtual geomagnetic pole and plotting it together with the North American plate apparent polar wander paths of *Besse and Courtillot* [2002] and *Torsvik et al*. [2001] shows remarkable coincidence with both (Figure 11a). The Manicouagan pole lies near the 205 Ma pole from the *Torsvik et al*. [2001] path and on the 210 to 220 Ma segment of the *Besse and Courtillot* [2002] path. The latter is in better agreement with the radiometric ages of the crater (e.g., 214 ± 1 Ma [*Hodych and Dunning*, 1992]).

Using an inclination of 41° to convert the 22.8 ± 7.7 μT paleointensity into a virtual dipole moment yields $4.9 \pm 1.6 \times 10^{22}$ Am^2. Compared to the global paleointensity database of *Biggin et al*. [2010] (Figure 11b),

the Manicouagan paleointensity result falls close to the average trend—again suggesting that secular variation at Manicouagan has been averaged out in our data set. In contrast, the virtual dipole moment obtained from the Upper Triassic Rochechouart (France) crater likely did not include enough time averaging [*Eitel et al*., 2014] and lies away from the average global trend (Figure 11b). Nonaveraging of secular variation of the Rochechouart data makes it difficult to use paleomagnetism (direction and intensity) to test the hypothesis for a fragmented, or Shoemaker-Levy 9-type, impact to explain the origin of both craters [*Spray et al*., 1998], although different magnetic polarities recorded in their impact melts suggest different formation times [*Kent*, 1998; *Eitel et al*., 2014].

Assuming an average thickness of 200 m for the impact melt at Manicouagan, *Onorato et al*. [1978] calculated that it took 5300 years to cool from 1227°C to 627°C. Knowing now that the impact melt can reach up to 1500 m in thickness [*Spray and Thompson*, 2008], one can also consider the cooling history of a 5 km wide by 1 km thick felsic magma chamber [*Bea*, 2010] as an analog for the thickest impact units near the center. Given an initial temperature of 1000°C, full crystallinity, as well as passage through the blocking temperature of hematite or magnetite, would not be reached even after 15,000 years in the center.

That only a single, normal magnetic polarity exists in all samples is not incompatible with the durations of the normal polarity chrons around 214 Ma (Late Triassic and Norian) that average ~1 million years (E13n and E14n in *Kent and Olsen* [1999] and Alaunian 2 in *Gallet et al*. [2007]). Following the heat conduction equations in *Turcotte and Schubert* [1982], and assuming that 100 m of impact melt has been eroded above the topographically highest sampling site, suggests that the uppermost site would take 382 years after impact to cool to 700°C, and 860 years to 500°C, while the lowest site would cool to 700°C 13,750 years after impact and to 500°C after 31,000 years. Given the large uncertainty surrounding the initial conditions and the exact cooling mechanism, we only wish to emphasize that the sampled stratigraphy spans from several hundred to several thousand or even tens of thousands of years following the impact. Anything above 3000–5000 years is within normal overturn times in the outer core, i.e., the time needed for a geomagnetic reversal [*Bogue and Merrill*, 1992; *Leonhardt and Fabian*, 2007; *Valet et al*., 2012], so we should have observed an effect had the Manicouagan impact perturbed the dynamo process.

We conclude that the seismic energy released from the Manicouagan impact was insufficient to disturb the dynamo process (such as through oscillation of the inner core) in a way that resulted in observable changes in field direction or intensity on the Earth's surface. Given the relatively large size of the Manicouagan crater together with the amount of time averaging allowed by the thick impact melt, it will be difficult to find a better target in which the impact-geodynamo hypothesis can be tested on Earth. It should be borne in mind, however, that while Manicouagan is relatively large in the context of Earth's limited impact crater inventory, it remains small by solar system standards [e.g., *Taylor*, 2001]. For example, the South Pole Aitken impact basin on the Moon is 2400 km in diameter, Hellas on Mars is 2300 km diameter, and Caloris on Mercury is 1550 km diameter. Larger impacts could have shock-heated the Martian core leading to the eventual demise of its dynamo [*Arkani-Hamed and Olson*, 2010] or changed the rotation rate of the Moon thereby generating a weak and short-lived lunar dynamo [*Le Bars et al*., 2011]. None of Earth's known surviving impact structures has a rim-to-rim diameter $>$300 km.

5. Conclusions

A comprehensive paleomagnetic and rock magnetic study of the Manicouagan impact melt sheet and uplifted basement rocks, combined with mineralogical, petrological, drilling, and field data, has yielded important constraints regarding the structure's formation:

1. There is a lack of a significant hydrothermal activity within the impact melt sheet. This is compatible with titanohematite developing at $>$570°C during oxi-exsolution as part of the melt crystallization process and implies that the impact event took place within a relatively dry setting on subaerial continental crust.
2. Magnetic anisotropy data, combined with field evidence, suggest that a record has, in places, been retained of the dynamic flow of impact melt during transient crater collapse and central uplift formation. It is difficult to constrain the sense of movement, but the direction is generally radial. The flow sense could be centrifugal (outward) or centripetal (inward). There is an indication that outward flow is favored, which, if correct, may be a response to the emergence of the central uplift during modification stage tectonics.

Large-scale convection likely plays a role several hundred meters below the surface in the thickest portion of the melt sheet.

3. Magnetic susceptibility data indicate that fractionation and titanomagnetite enrichment occurred in the lower portion of impact melt that formed in a trough structure that accumulated over 1 km of impact melt, which may be a fault-bounded graben, as suggested by *Spray and Thompson* [2008]. The central magnetic anomaly high is partially produced from the titanomagnetite-enriched deeper levels in the thick impact melt; however, meta-gabbroic rocks likely account for the bulk of the anomaly.
4. The central, thicker impact melt unit does not record any change in magnetic field direction or intensity. This indicates that the Manicouagan impact was not large enough to affect Earth's dynamo, and in combination with our previous work [*Koch et al*., 2012; *Eitel et al*., 2014; *Hervé et al*., 2015], any link between meteorite impacts and geodynamo disturbances should be considered improbable, at least during the Phanerozoic when impact frequency and meteorite sizes were lower than during the earlier stages of Earth's evolution [*French*, 1998]. However, it should be kept in mind that the Manicouagan event is relatively small at the solar system scale, e.g., the South Pole Aitken crater on the Moon has a diameter of 2400 km versus Manicouagan's 85 km.

Acknowledgments

Funding for field and laboratory work was provided by the Deutsche Forschungsgemeinschaft to S.A.G. (project GI712/6-1) under the auspices of SPP1488, Planetary Magnetism. Field operations were also supported through the University of New Brunswick's *Manicouagan Impact Research Program*, which is funded by grants awarded to J.G.S. from the Canadian Space Agency, the Canada Research Chairs program, and the Natural Sciences and Engineering Research Council of Canada. We thank Nicolas Swanson-Hysell, Mike Dence, and an anonymous reviewer for their helpful suggestions that improved this work. Editorial handling by André Revil and an anonymous Associate Editor are appreciated. Any data herein can be obtained by contacting the corresponding author.

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